

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Manipulation (BL2)	Assessment:	Assessment Tool 1 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10Marks)
CO1 Statement:	Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)		
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Section / Batch:	AI&DS 2024/2028	Date:	21/07/28

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Code of Conduct
 I Muzammil - 192424279 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Muzammil

- Instructions**
- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
 - No negative marking. Unanswered questions carry zero marks.
 - No calculators or electronic devices permitted.
 - Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Aesthetic Mapping & Dashboard Design	Mapping categorical, ordinal, and quantitative data to position, size, shape, and color; selection of scales, axes, and coordinate systems for business dashboards	1	10
Nonlinear Axes & Health Data Visualization	Use of logarithmic scales for exponential data, time-series representation, coordinate systems, and color scales for highlighting high-risk regions	2	10
GRAND TOTAL:			20 Marks

Case study: Multinational Retail Company Dashboard

Company Background:

A multinational retail company has hundreds of stores located in different regions.

The company collects the following data every month:

Data collected:

Stores Region

Product Category

Monthly sales

Customer Satisfaction

Profit margin

Promotional Campaign

Types of Data:

Categorical.

Categorical.

Quantitative

Ordinal

Quantitative

Ordinal.

Current problem:

- * Which region performs best
- * Which products used the most
- * Which stores have low customer satisfaction

Objective:

- * Compare store performance
- * Monitor monthly sales
- * Identify top-selling products
- * Evaluate customer satisfaction.
- * Detect low-performing stores quickly.

Need-to-explain:-

1. Data Mapping:

which visual element should represent each type of data?

Data

Visual Mapping

Product category

Sales

Profit margin

Customer rating

Region

color

Bar height

Bubble size

colour integrity

Position on dashboard

2. Visualization Selection

- * Bar chart $\rightarrow$ product sales
- * Line chart $\rightarrow$ Monthly sales trend
- * Pie chart $\rightarrow$ Category share
- * Scatter plot $\rightarrow$ Sales vs profit
- * Heat map $\rightarrow$ Regional performance
- * KPI cards $\rightarrow$ Total sales profit

3) Scales and Axes

- * Linear scales for sales

x-axis = Months

y-axis = Revenue

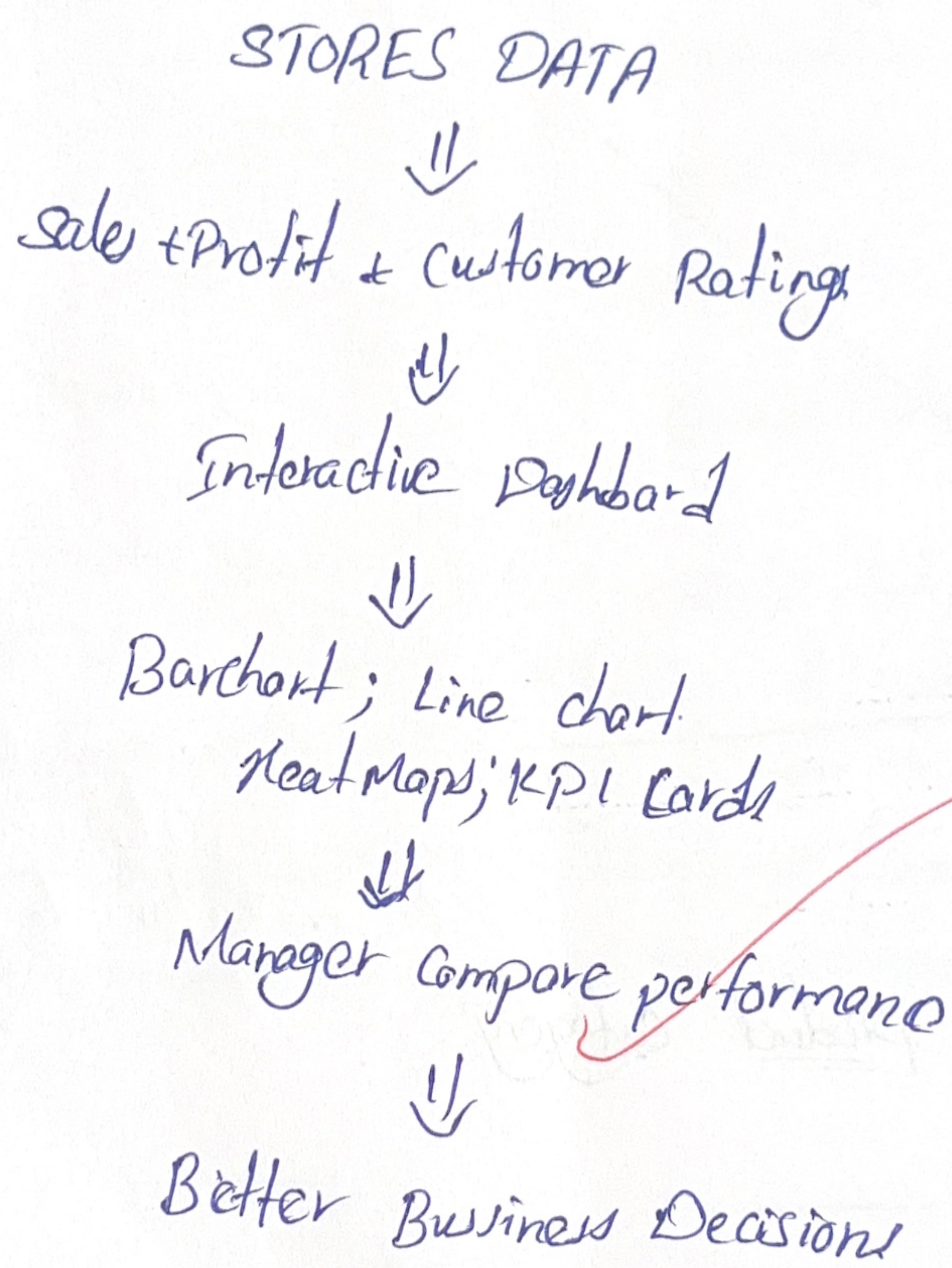
- * Co-ordinate system = Cartesian Co-ordinate

- Color Usage
- Differentiate product categories.
- Highlight high-performing stores
- show low-performing stores in red
- use green for good performance
- use yellow for average

Justification:

Explain why these charts & colours help managers make faster & better business decisions

Flow chart:

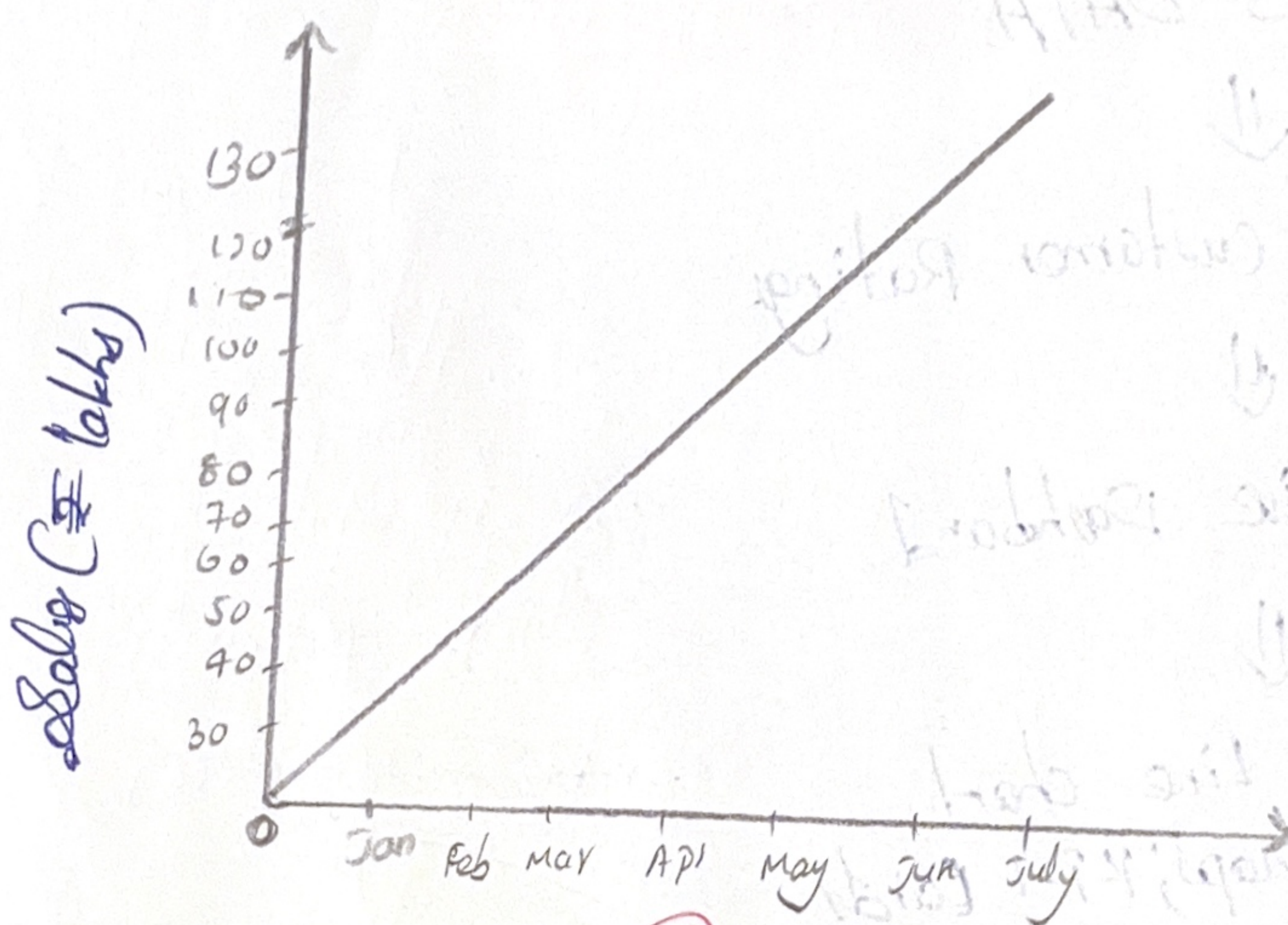


Conclusion:

An interactive retail analytics dashboard enables managers to effectively monitor sales performance, customer satisfaction, profit margins & promotional campaign success across different regions. By mapping categorical, ordinal & quantitative data to appropriate visual elements such as color, size & position the dashboard presents information in a way.

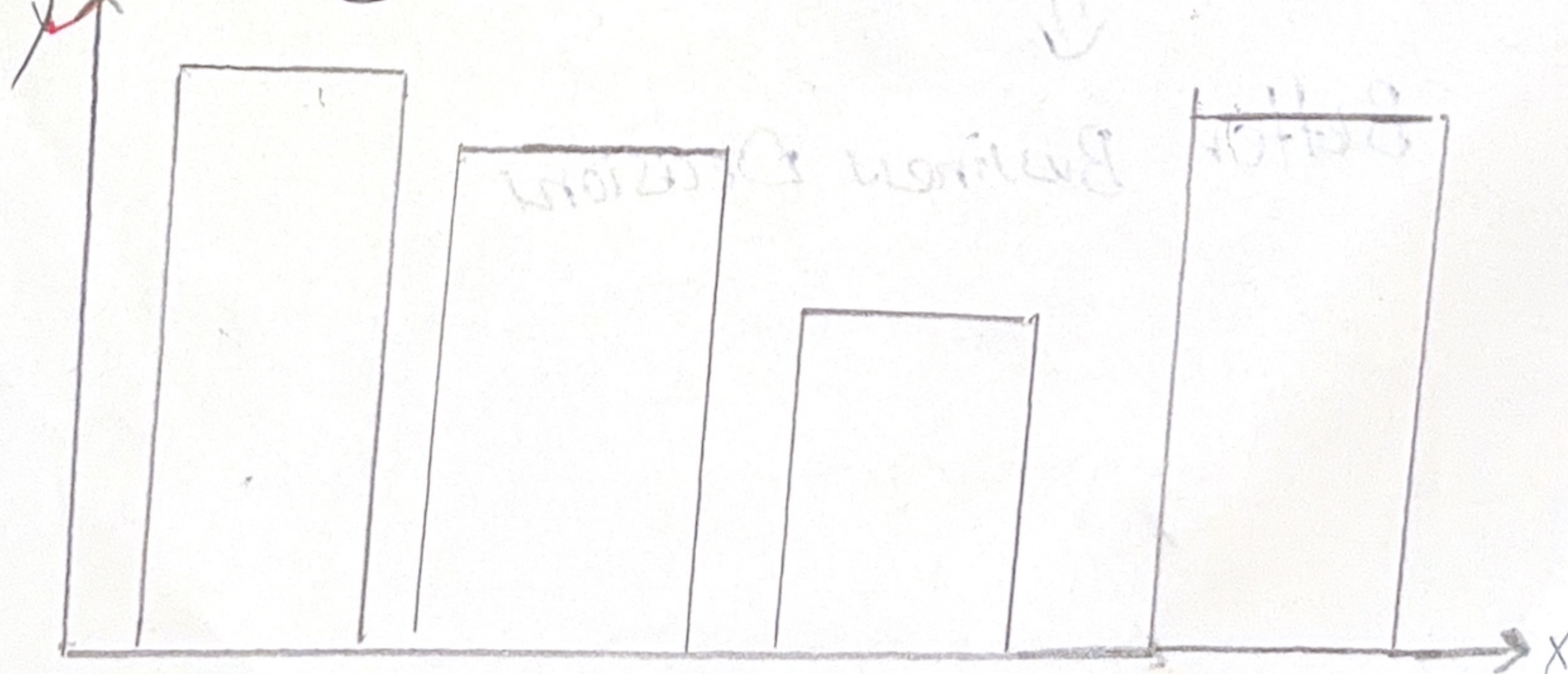
Line chart:

Monthly sales trend:



Bar chart:

Monthly sales by product category:



Case study: National Health Agency

Organized - Background:

A national health agency monitors the spread of a contagious disease across many state hospitals and daily reports.

Data Collected:

DATA	TYPE
Daily infection count	Quantitative
Mortality rate.	Quantitative
Mortality rate	Quantitative
Vaccination coverage	Quantitative
Population density	Quantitative
Disease Growth rate	Quantitative

⇒ The national agency (Health) collects data from the hospitals & health care centers every day to monitor the spread of a contagious disease

⇒ This data helps officials understand the current situation, identify high-risk areas of more informed decisions for diseases.

Current problem:

Some states experience a very rapid increase in infection

DAY	CASES.
Day 1	50
Day 2	100
Day 3	250
Day 4	700
Day 5	1800

If plotted on a normal graph, smaller values become difficult to see.

Objective: officials need a visualization system to

- * Identify diseases hotspot
- * Monitor daily spread
- * Compare states

⇒ visualization strategy enables health officials to quickly identify outbreak hotspot monitor diseases progression.

Mapping

map each variable to an appropriate visual element

Data

State

Infection count

Growth rate

Mortality rate

Vaccination coverage

Visual Mapping -

map position

circle size

line height

colour intensity

Green colour.

Visualization selection:

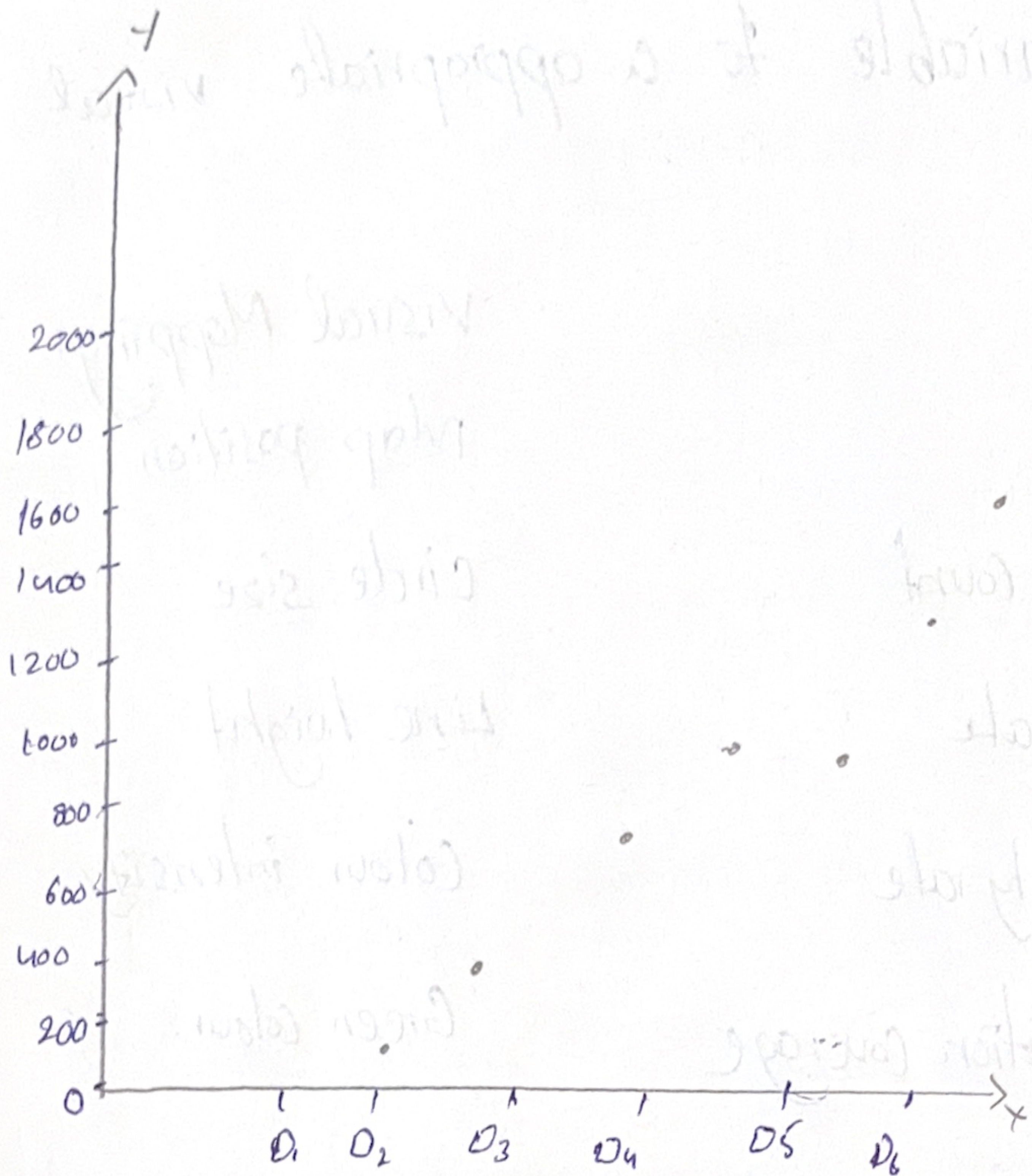
* Line chart → Daily - Infection Trend

* Choropleth Map → state-wise risk

* Bubble map → No. of cases.



Visualization 1: Line Graph: Daily Infection trend



Bar chart: Vaccination coverage

